CSCI 152 Performance and Data Structures
Programming Assignment 1

Deadline: Sunday, June 15, 2025 at 11:59 PM

1 DynamicMatrix

This task provides an implementation of the `DynamicMatrix` data structure, defined in the header file `DynamicMatrix.h`. In particular, `DynamicMatrix` is a class template that represents a 2D array (matrix) of dynamic size capable of storing any data type. It maintains the number of rows and columns and supports dynamic resizing and matrix operations such as transpose.

You are given `DynamicMatrix.h`, which defines the interface and member declarations of the class. You are also provided with `DynamicMatrix.cpp`, which contains function implementations of the class methods. You are required to complete and understand the implementation file `DynamicMatrix.cpp`. A sample driver file `main.cpp` and a `Makefile` are also provided for building and testing your implementation.

Before you begin, please review the class policies regarding programming assignments and lab exercises. **No live grading is required** for this assignment.

2 Tasks

1. Default constructor initializes a 2x2 matrix of default values.
2. Parameterized constructor allows specifying the number of rows and columns.
3. Copy constructor and assignment operator ensure deep copies of matrix data.
4. Overloaded `operator()(size_t i, size_t j)` to access elements at position `(i, j)`.
5. `size_t numRows() const` – returns the number of rows in the matrix.
6. `size_t numCols() const` – returns the number of columns in the matrix.
7. Destructor releases dynamically allocated memory.
8. `fill(const T& value)` – fills the matrix with the provided value.
9. `resize(size_t newRows, size_t newCols)` – resizes the matrix while preserving existing data as much as possible.

10. `transpose()` – returns a new matrix that is the transpose of the current matrix.
11. `print(std::ostream& out) const` – prints matrix in row-column format.
12. Overload `operator<<` for printing the matrix to output streams.

3 Submit

Upload the following files to Moodle under "Programming Assignment 1" by the deadline:

- `DynamicMatrix.cpp`
- `DynamicMatrix.h`
- `main.cpp`
- `Makefile`

The names and file extensions must be exact and should not be archived. All class names and function signatures must match the provided definitions exactly. Failure to do so may result in your code not being recognized by the testing system, which will result in a score of 0.

4 Questions?

If you have questions about the assignment, please ask during lab sessions or post your questions on Piazza. Do **not** post your code publicly. Instead, post your questions as "private to instructors" on Piazza.